

Aman Sharma

 Github |  LinkedIn |  sharmaaman9058@gmail.com |  +91-9058490490

EDUCATION

Indraprastha Institute of Information Technology, Delhi M.Tech. CSE	Aug 2024 - May 2026 CGPA: 7.4/10.0
Gurukula Kangri University, Haridwar, Uttarakhand B.Tech. CSE	2020 - 2024 CGPA: 8.4/10.0
Sanskriti Public School, Baldeo, Mathura CBSE Standard 12th, PCM	2018 - 2019 Percentage: 76.8%
Sanskriti Public School, Baldeo, Mathura CBSE Standard 10th	2016 - 2017 Percentage: 64.6%

EXPERIENCE

MIDAS Lab, IIIT-Delhi Postgraduate Researcher (Supervisor: Dr. Rajiv Ratn Shah)	Dec'2024 - May'2026 Delhi, India
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Tech Stack: Python, OpenCV, PyTorch, YOLOv8, AMD Ryzen 5 4600H CPU, NVIDIA GeForce GTX 1650 Ti GPU, dual 4K cameras, distance-measuring device.

Description: **OptiRSDE** is a real-time stereo depth estimation pipeline that integrates optimized feature matching, QR-based synchronization, and temporal smoothing to achieve <3% error at 50m and near 7.5% at 100m, running 10-20x faster than conventional methods for robotics, autonomous driving, and industrial safety.

CodeClause Data Science Intern	Mar 2023 - Apr 2023 Remote/India
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Tech Stack: Python, NumPy, Pandas, scikit-learn, Matplotlib, Plotly, Jupyter Notebook.

Description: During my internship, I developed a stock price prediction model for Google data from 2016 to 2021 using linear regression. Achieved an R^2 score of 0.78 on training and 0.80 on testing data, with MSE of 32,790.55 (train) and 26,360.56 (test). The project involved data preprocessing, feature scaling, regression modeling, and interactive visualization using Plotly.

PROJECTS

Student Learning Pathway Explorer & Tracker (Agentic AI) Supervisor: Dr. Vinayak Abrol	Jan 2026 - Apr 2026 
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Tech Stack: Python, Streamlit, LangChain, Hugging Face Sentence-Transformers, FAISS, Open-source LLMs/LLM API, OCR, Neo4j, Docker, RAG Pipeline, Git & GitHub.

Description: This project is an offline academic and career advisory system for M.Tech students that unifies course records, transcripts, and resumes into a structured format to provide personalized guidance. The implementation integrates specialized agents with a Neo4j knowledge graph to map relationships between skills and placements, while an offline-first RAG pipeline handles institutional regulations. So far, the multi-agent workflow has been successfully developed to process student profiles and suggest evidence-based academic pathways while ensuring complete user privacy.

RAG-Powered Assistant HCL Hackathon Project	Nov 2025 
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Tech Stack: Python, Streamlit, LangChain, Hugging Face Sentence-Transformers, FAISS, Open-source LLMs/LLM API, PyPDF2/pypdf, Document Chunking, RAG Pipeline, Streamlit Cloud Deployment, Git & GitHub.

Description: Built and deployed a RAG-based Document Assistant with PDF/text ingestion, intelligent chunking, MiniLM-L6-v2 embeddings, and FAISS semantic search to enable accurate context-aware responses. Developed an interactive Streamlit UI for document upload and chat-based querying, and deployed the full system on Streamlit Cloud with a scalable modular architecture.

Mental Health Memes Classification Supervisor: Dr. Md Shad Akhtar	Feb 2025 - Apr 2025 
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Tech Stack: Google Colab, Python, MentalBERT, PaddleOCR, RAG, Sentence Transformer, LLaVA.

Description: Implemented a multimodal system to detect depression (multi-label) and anxiety (single-label) symptoms in memes using OCR text, visual cues, and figurative reasoning. Utilized Sentence-BERT, prompt-based LLMs, and RAG-based knowledge fusion to generate contextual embeddings. Evaluated on custom-labeled datasets, resulting in Macro-F1 0.62 and Weighted-F1 0.61, surpassing baseline models.

A Virtualized NL2SQL System for Distributed Job Recommendation Supervisor: Dr. Mukesh Mohania	Aug 2025 - Nov 2025 
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Tech Stack: Llama-3 (via Groq), Gemini-Flash, Python, MySQL, Regex, JSON.

Description: The VirtuMatch project is a Federated NL2SQL System that queries distributed, independent MySQL databases using LLMs (Llama-3/Gemini-Flash). It features an AI-driven ETL module to dynamically merge heterogeneous schemas and uses Regex fallback for resilient execution. It provides a unified, ranked, and summarized answer.

ACHIEVEMENTS

Amazon ML	Under 800 rank out of 82000+ participants.	
GATE 2024	Qualified with rank 4610, Score - 523.	2024
JEE Main 2020	92.33th Percentile.	2020
LeetCode	Completed 100+ questions.	
Certification	National Conference on Knowledge, Cultural Traditions & Practices of India.	
Certification	COMMUNICODE (HackerRank) participation in Code-Decode.	
Coding Contest	Secured 4th Position organized by CHARABANC CLUB GKV.	

TECHNICAL SKILLS

Languages: Python, MATLAB, SQL, C++, NumPy, Pandas, Matplotlib
AI/ML: NLP, Machine Learning, Deep Learning, LLM, RAG, Computer Vision
Frameworks: PyTorch, HuggingFace
Tools: Kaggle, Google Colab, Jupyter Notebook, Figma, Git, GitHub
Expertise: Data Structures and Algorithms, OOP, Linux

POSITIONS OF RESPONSIBILITY

Teaching Assistant, IIIT-Delhi: Assisted in Data Structures & Algorithms (DSA), Environmental Studies (EVS), and Research Methodology courses.	Aug 2024 - May 2026
Student Mentor, IIIT-Delhi: Selected as a mentor to guide and support the incoming M.Tech 2025 batch.	Jul 2025 - Aug 2025
Captain, Team Shadow: Led Team Shadow to victory as Captain in the Indraprastha Premier League.	Apr 2026 - May 2026